

CountUpTimer

With the `CountUpTimer` class, you create a timer object and tell it when to start. Then, in every frame, you can call one of three different methods to get the time elapsed in different formats.

[Listing 13-4](#) contains the implementation of the `CountUpTimer` class from `pyghelpers`. The code is a good example of how the different methods of a class share instance variables.

File: (Available as part of the `pyghelpers` module)

```
# CountUpTimer class

class CountUpTimer():
    --- snip ---

    def __init__(self): ❶
        self.running = False
        self.savedSecondsElapsed = 0.0
        self.secondsStart = 0 # safeguard

    def start(self): ❷
        --- snip ---
        self.secondsStart = time.time() # get the current seconds and save the
value
        self.running = True
        self.savedSecondsElapsed = 0.0

    def getTime(self): ❸
        """Returns the time elapsed as a float"""
        if not self.running:
            return self.savedSecondsElapsed # do nothing

        self.savedSecondsElapsed = time.time() - self.secondsStart
        return self.savedSecondsElapsed # returns a float

    def getTimeInSeconds(self): ❹
        """Returns the time elapsed as an integer number of seconds"""
        nSeconds = int(self.getTime())
        return nSeconds

# Updated version using fStrings
```